

Quadratic placement

Quadratic-lite placement averages neighbors under fixed pads, here A and D stay pinned

The idea

- Gauss–Seidel style
- Fixed pads anchor the system
- Teaching point: pad constraints raise HPWL versus unconstrained force on the same seed

Pseudocode

- Quadratic-lite solves free cells toward neighbor averages under pinned pads
- Damping keeps the system from collapsing
- Open this module's examples file and find the Pseudocode section
- That written sketch is what you implement on the implement track and what the browser

Algorithm sketch

- With A and D fixed

Algorithm sketch — try these

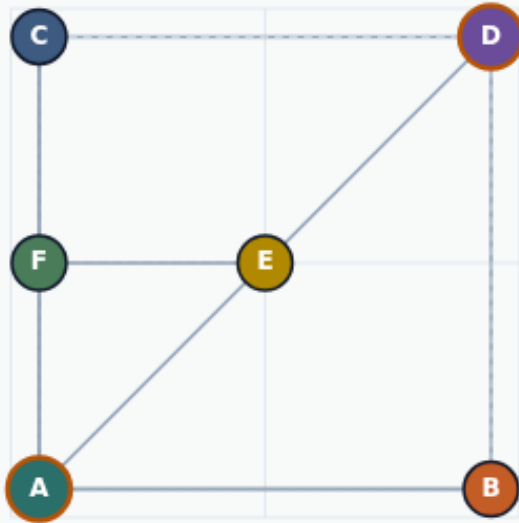
```
INPUT: positions, nets, fixed pads {A,D}  
OUTPUT: free-cell coords + HPWL  
repeat: for free c:  
    blend toward neighbor average (damped)  
pads A,D remain pinned  
GOLDEN starter 52 → HPWL 48
```

Pads A and D stay fixed

QUADRATIC PLACE

Pads A and D stay fixed

Step 1 / 5 · pads-fixed · module02-03-quadratic-place



Explanation

Quadratic-lite averages neighbors under fixed pads. Here A stays at (0,0) and D at (8,8)—anchors that free cells must respect.

- Fixed pads: A, D
- Gauss–Seidel neighbor average
- Blend retains some prior coord

HPWL: 52

A @ (0,0) · D @ (8,8)

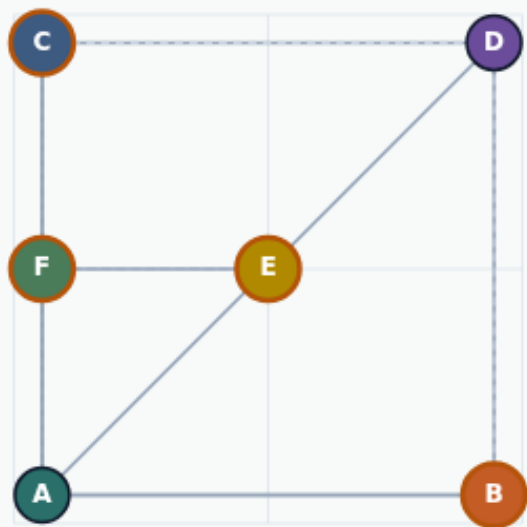
Pads A and D stay fixed

Free cells slide toward neighbors

QUADRATIC PLACE

Free cells slide toward neighbors

Step 2 / 5 · free-cells · module02-03-quadratic-place



Explanation

B, C, E, and F update toward the average of their neighbors while A and D hold. The system cannot collapse onto the pads.

- Update only non-pad cells
- blend ≈ 0.55 toward average
- Pads raise HPWL vs free force

Free: B, C, E, F
Fixed: A, D

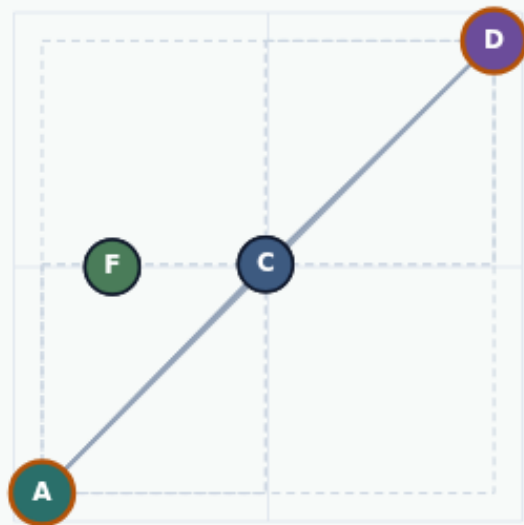
Free cells slide toward neighbors

After quadratic: HPWL forty-eight

QUADRATIC PLACE

After quadratic: HPWL forty-eight

Step 3 / 5 · after-quad · module02-03-quadratic-place



Explanation

Default iterations reach total HPWL forty-eight. Pads constrain the free cells, so you will not match free force's eighteen point seven.

- Starter 52 → 48
- Pads keep the spread
- Teaching point: constraints cost WL

Before: 52
After: 48
Measured: 48

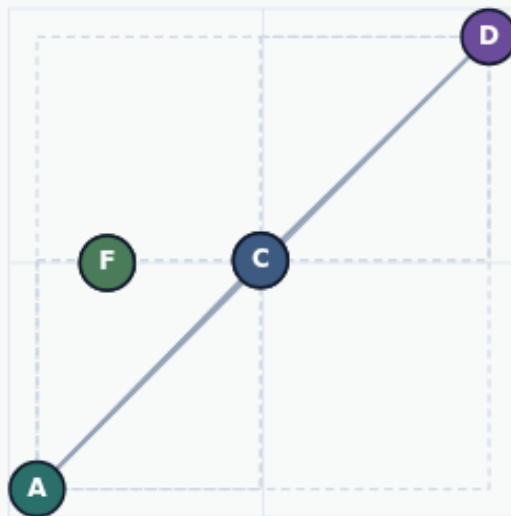
After quadratic: HPWL forty-eight

Quadratic vs unconstrained force

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Quadratic vs unconstrained force

Step 4 / 5 · vs-force · module02-03-quadratic-place



Explanation

Force without pads reaches about eighteen point seven; quadratic with pads stays at forty-eight. Same seed, different constraint story.

- Force (free): ≈ 18.7
- Quadratic (pads): 48
- Both beat starter 52

```
forceHpwlAfter: 18.7  
quadraticHpwlAfter: 48
```

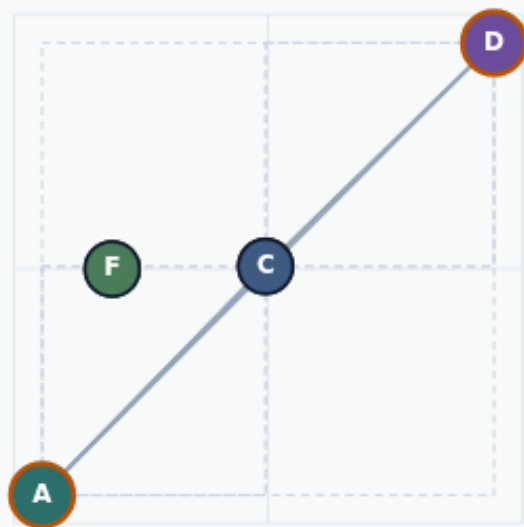
Quadratic vs unconstrained force

Pads anchor the solve

QUADRATIC PLACE

Pads anchor the solve

Step 5 / 5 · takeaway · module02-03-quadratic-place



Explanation

Always declare fixed cells. On this instance the quadratic golden is forty-eight—modest wirelength win, honest about pad constraints.

- Fixed A, D
- Starter 52 → 48
- Next: analytical density stage

quadraticHpwlAfter: 48

Pads anchor the solve

Browser lab track

- In the browser lab track, open the **quadratic-place** lab from the tools shelf
- Load the starter placement, run the algorithm once
- Work the challenges that lock the goldens

Implement track

- In the implement track
- Parse ``tiny_place.json``, run the algorithm with a deterministic seed
- Match the browser goldens before you claim the checklist

Pitfalls

- Common traps

Your turn

- Complete the checklist for at least one track, preferably both
- Implement until your metrics match the starter goldens
- When you're ready, take the short quiz, then continue to the next module